

國立臺灣師範大學
理學院
地球科學系
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College of Science
National Taiwan Normal University
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原生恆星盤外圍絲狀結構之模型與觀測比較
Modeling Streamer Structures in Collapsing
Prestellar Cores

莊宗翰
Tsung-Han Chuang

指導教授：李悅寧 博士
Advisor: Yueh-Ning Lee, Ph.D.

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摘要

Streamers have been observed with high-resolution ALMA observations around many protoplanetary disks undergoing formation. They have been suggested to dominate the mass accretion budget from the collapsing prestellar core, and have important consequences on the disk dynamics. Understanding the formation of streamers is therefore important for explaining how protoplanetary disks receive mass from the envelope. We propose a model to explain the formation of streamer structures by considering density enhancements due to gravitational instability. We test our model against sources where streamers have been detected (ex., Per-emb-2 and Per-emb-50 observed with NOEMA) and fit for model parameters. This allows us to gain deeper insights into the physical origin of streamers and their role in mass transport from the core to the disk. This model can be applied to analyze new observations as well as archival data that show signs of streamers.

Abstract

Streamers have been observed with high-resolution ALMA observations around many protoplanetary disks undergoing formation. They have been suggested to dominate the mass accretion budget from the collapsing prestellar core, and have important consequences on the disk dynamics. Understanding the formation of streamers is therefore important for explaining how protoplanetary disks receive mass from the envelope. We propose a model to explain the formation of streamer structures by considering density enhancements due to gravitational instability. We test our model against sources where streamers have been detected (ex., Per-emb-2 and Per-emb-50 observed with NOEMA) and fit for model parameters. This allows us to gain deeper insights into the physical origin of streamers and their role in mass transport from the core to the disk. This model can be applied to analyze new observations as well as archival data that show signs of streamers.

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Chapter 1

Introduction

1.1 The formation of cores

Stars form within cold, dense regions of molecular clouds, where local density enhancements can initiate gravitational collapse. Once a sufficiently overdense region begins to contract, it forms a dense core, a gravitationally bound structure that represents the earliest stage of star formation. Before any central heating source develops, the collapsing dense core is referred to as a prestellar core, characterized by low temperatures (~ 10 K) and the absence of an internal heat source.

As the collapse proceeds and densities rise, the gas eventually becomes optically thick to its own radiation. At this point, radiative cooling becomes inefficient, and the collapse slows as the pressure begins to increase adiabatically. This transition leads to the formation of the first hydrostatic core (FHSC), which is a short-lived structure supported primarily by adiabatic pressure. The FHSC phase lasts only a few thousand years and marks a key evolutionary stage between the prestellar core and the protostellar core.

When the temperature in the center increases to ~ 2000 K, molecular hydrogen begins to dissociate, a process that absorbs energy and reduces thermal pressure support. This triggers a second, rapid collapse and gives rise to the protostar,

the central object that continues to grow by accreting material from its surrounding envelope. Although accretion from the envelope begins during the prestellar and FHSC stages, the protostar phase formally begins once the second collapse produces a gravitationally bound central object.

The envelope surrounding the forming protostar is a cold, dusty reservoir of infalling material. It typically extends over thousands to tens of thousands of AU, and in the earliest stages the envelope retains a large fraction of the system’s mass.

During the early Class 0 and Class I stages, the protostar remains enshrouded within this parental envelope. In this turbulent and gravitationally unstable environment, small perturbations within the envelope can grow nonlinearly, giving rise to asymmetric, filamentary overdensities known as streamers. These streamers represent dense structures within the infalling envelope and, unlike the rotationally supported disk, do not necessarily follow Keplerian motion. They are increasingly recognized as an important channel for transporting material from the envelope onto the disk, thereby influencing disk growth, stability, and angular-momentum redistribution.

Over time, as accretion weakens and the envelope dissipates, the protostar evolves into a pre-main-sequence star, eventually contracting toward the main sequence where sustained hydrogen fusion begins.

1.2 The streamer in observations

Recently, several observations revealed asymmetric structures in the protostellar system (e.g., Pineda et al. 2020, Valdivia-Mena et al. 2022, Hsieh et al. 2023). The features of streamers are very diverse in morphology, do not always form in a disk plane, and have very weak density enhancement in observation.

1.3 Current Limitations and Our Approach

1.4 Overview of Targets Used in This Study

1.5 Our work overview

The formation process of protoplanetary disks is still unrefined. Given their importance, we have developed a new model to characterize the streamer features and kinematics, focusing on gravitationally driven infalling and rotation.

Chapter 2

Methods

2.1 Observational Data

To test our model against protostellar systems in which streamer-like structures have been reported, we use five observational datasets: Per-emb-2, Per-emb-50, S CrA, HL Tau, and SVS 13A. Per-emb-2 and Per-emb-50 were observed with the Northern Extended Millimeter Array (NOEMA), whereas S CrA, HL Tau, and SVS 13A were observed with the Atacama Large Millimeter/submillimeter Array (ALMA). For Per-emb-50, S CrA, HL Tau, and SVS 13A, we extract the position-position-velocity (PPV) information of the streamer directly from the calibrated data cubes, while for Per-emb-2 we work with the moment maps. A summary of the observational properties of all datasets is provided in Table 2.1.

Per-emb-2 was observed in D-configuration on June 22, July 27–28, and August 19, 2018, and in C-configuration on October 9, 10, 17, and 18, 2018 (project S18AG; (Pineda et al., 2020)). The baselines ranged from 16.4 to 287.8 m.

Per-emb-50 was observed in C-configuration on December 29, 2019, and January 5, 2021, and in D-configuration on August 6, September 7, and 8, 2020 (Valdivia-Mena et al., 2022). The synthesized beam of the cube used in this work is $1''.24 \times 0''.72$.

SVS 13A was observed with ALMA in Band 6, producing the DCN ($J = 3-2$) data cube used in this study. The cube data has a synthesized beam of $0''.15 \times 0''.11$.

The S CrA dataset was obtained from the ALMA Science Archive and was observed on November 6, 2019 (project 2019.1.01792.S; (Gupta et al., 2023)). The data cube has a spatial sampling of $0''.05$ per pixel.

HL Tau was observed with ALMA in the C40-7 configuration on August 22, 28, and September 1, 2017, with a maximum projected baseline of ~ 3.3 km (Yen et al., 2019). The restoring beam of the HCO^+ data cube used in this work is $0''.10 \times 0''.09$.

2.1.1 Per-emb-2

Per-emb-2 (IRAS 03292+3039) is a young Class 0 protostellar object in the Perseus molecular cloud at a distance of roughly 300 pc (Zucker et al., 2018). The source contains a close binary with a separation below 20 au and show a clear outflow in CO (2-1) (Stephens et al., 2019). NOEMA imaging has revealed a streamer structure connecting scales of more than 10,000 au to within a few hundred au of the protostar (Pineda et al., 2020).

2.1.2 Per-emb-50

Per-emb-50 is a Class I protostar in the NGC 1333 region at a distance of ~ 293 pc (Ortiz-León et al., 2018). It is relatively luminous among sources of similar evolutionary stage and hosts a compact disk with a characteristic radius of ~ 30 au (Segura-Cox et al., 2016). Molecular outflow activity is observed roughly east–west (Stephens et al., 2019). The velocity structure along the streamer shows a gradual increase in blueshift toward smaller radii.

2.1.3 S CrA

2.1.4 HL Tau

2.1.5 SVS 13A

Table 2.1: Summary of the observational datasets used in this work.

Target	Telescope	RA (J2000)	DEC (J2000)	Line	Beam (")
Per-emb-2	NOEMA	03:32:17.932	+30:49:47.705	HC ₃ N (10–9)	3".73×3".02
Per-emb-50	NOEMA	03:29:07.768	+31:21:57.128	C ¹⁸ O (2–1)	1".24×0".72
SVS 13A	ALMA	03:29:03.740	+31:16:03.800	DCN (3–2)	0".15×0".11
S CrA	ALMA	19:01:08.600	−36:57:20.000	¹³ CO (2–1)	—
HL Tau	ALMA	04:31:38.428	+18:13:57.190	HCO ⁺ (3–2)	0".10×0".09

2.2 Build up a model

We build a model that can reproduce the observed streamer by considering a self-collapse density enhancement structure with a rotation velocity and evolve with time in 3D space.

2.3 Formalism

In this section, we summarize the formalism of the model, including the density enhancement along the polar angle (θ), the infall velocity (v_{inf}), the rotation velocity (v_{rot}), the accumulated rotation (ϕ), and the inclination (i). The sonic length.

$$r_s = c_s T \quad (2.1)$$

Infall velocity

$$v_{\text{inf}} = -\sqrt{\frac{2GM_*}{r}} - 3.3c_s \quad (c_s = 200 \text{ m s}^{-1}) \quad (2.2)$$

The rotation velocity.

$$v_{\text{rot}} = r_s \omega \Omega_{\text{kep}} \left[\left(\frac{r}{r_s} \right)^{-1} + \left(\frac{r}{r_s} \right)^{-\frac{1}{3}} \right] \quad (2.3)$$

The accumulated rotation.

$$\phi = \phi_0 + t_s \omega \Omega_{\text{kep}} \sqrt{\frac{2c_s^3 t_s}{GM_*}} \left[\left(\frac{r}{r_s} \right)^{-\frac{1}{2}} + \left(\frac{r}{r_s} \right)^{-\frac{1}{3}} \right] \quad (2.4)$$

2.4 Coordinate transformation

Position

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases} \quad (2.5)$$

Infall velocity

$$\begin{cases} v_{x_{\text{inf}}} = v_{\text{inf}} \sin \theta \cos \phi \\ v_{y_{\text{inf}}} = v_{\text{inf}} \sin \theta \sin \phi \\ v_{z_{\text{inf}}} = v_{\text{inf}} \cos \theta \end{cases} \quad (2.6)$$

Rotation velocity

$$\begin{cases} v_x = -v_{\text{rot}} \sin \theta \sin \phi + v_{x_{\text{inf}}} \\ v_y = v_{\text{rot}} \sin \theta \cos \phi + v_{y_{\text{inf}}} \\ v_z = 0 + v_{z_{\text{inf}}} \end{cases} \quad (2.7)$$

2.5 Streamer on the sky in the PPV space

Position rotates by inclination. Inclination(i): backward (+), forward (-).

$$\begin{cases} x' = r \sin \theta \cos \phi \\ y' = y \cos i + z \sin i \\ z' = -y \sin i + z \cos i = r(-\sin \theta \sin \phi \sin i + \cos \theta \cos i) \end{cases} \quad (2.8)$$

Velocity rotates by inclination

$$\begin{cases} v'_x = v_x \\ v'_y = v_y \cos i + v_z \sin i = v_{\text{rot}} \sin \theta \cos \phi \cos i + v_{\text{inf}}(\sin \theta \sin \phi \cos i + \cos \theta \sin i) \\ v'_z = -v_y \sin i + v_z \cos i = -v_{\text{rot}} \sin \theta \cos \phi \sin i + v_{\text{inf}}(-\sin \theta \sin \phi \sin i + \cos \theta \cos i) \end{cases} \quad (2.9)$$

θ corresponds to the initial polar angle of the streamer, ϕ_0 is the initial azimuth angle, i is the inclination angle (zero inclination corresponds to an edge-on disk), ω is the ratio of Keplerian rotation velocity, and t_s is the collapsing time of the particle within the streamer.

2.6 Fitting procedure

The fitting procedure described below is applied to each dataset to constrain the model parameters. The results are compared to the observed streamer morphology

and kinematics to assess the validity of the model.

2.6.1 Position angle

[(YN) This step can be neglected if the disk PA on the sky is known. Otherwise, the previous step needs to be repeated until the optimal is found.]

2.6.2 Extracting the position and velocity from observations

To characterize the observed streamer morphology, we extract position and velocity centroids as a function of projected radius from the masked data cube. The procedure consists of two stages. First, we compute preliminary centroids in a set of radial bins using an intensity-weighted average with a coarse directional selection. A hand-traced streamline provides an approximate $\theta(r)$ relation, from which a preliminary centroid direction and a spatial dispersion are determined for each radial interval. These preliminary centroids are then used to construct a smooth reference angle profile $\theta_{\text{ref}}(r)$ and an angular dispersion $\sigma_{\theta}(r)$, obtained by converting the local centroid scatter in (x, z) to an angular width.

In the second stage, we recompute the centroids using Gaussian directional weights that favor pixels aligned with the local streamer direction. For each radial bin, the weight is defined as

$$w_{\theta} = \exp \left[-\frac{(\theta - \theta_{\text{ref}}(r))^2}{2\sigma_{\theta}(r)^2} \right],$$

where θ is the polar angle of each pixel in the image plane. The intensity-weighted

centroid position is then computed as

$$\bar{x} = \frac{\sum I w_\theta x}{\sum I w_\theta}, \quad \bar{z} = \frac{\sum I w_\theta z}{\sum I w_\theta},$$

and the velocity centroid is obtained from

$$\bar{v} = \frac{\sum I w_\theta v}{\sum I w_\theta},$$

where v is the channel velocity converted into the LSR frame. This two-stage procedure produces a set of smooth, noise-suppressed position-velocity centroids that faithfully trace the observed streamer trajectory. The number of centroids extracted for each target is summarized in Table 2.2.

2.6.3 Grid search parameter estimation

We model the streamer kinematics using five parameters $(\theta, \phi_0, i, t_s, \omega)$, corresponding to the initial polar and azimuthal angles of the streamer, the inclination of the system, the characteristic collapse time, and the ratio of rotation velocity. For each target, we first perform a coarse five-dimensional grid search to identify reasonable regions of parameter space. The grid covers $\theta \in (0, \pi/2)$, $\phi \in (0, 2\pi)$, and $i \in (-\pi/2, \pi/2)$ with uniform sampling, while the collapse time t_s is the characteristic time after the central mass singularity formation, and ω is sampled linearly between 0 and 1.

The grid search uses the extracted position-velocity centroids to evaluate the model performance. At each grid point, we compute the misfit between the model streamer and the observed centroids using the error metric

$$E = \sqrt{\langle \Delta x^2 + \Delta z^2 + w_v \Delta v^2 \rangle},$$

where Δx and Δz are the projected position differences (in AU), and Δv is the line-of-sight velocity residual (in km s^{-1}). The weight

$$w_v = \left(\frac{\Delta x_{\text{AU}}}{\Delta v_{\text{km s}^{-1}}} \right)^2$$

is chosen such that spatial and velocity deviations contribute in comparable physical units.

2.6.4 Markov Chain Monte Carlo exploration

After constructing data-driven prior ranges from the grid search, we perform Markov Chain Monte Carlo (MCMC) exploration of the five parameters. The MCMC sampling is carried out using the `emcee` package, which implements an affine-invariant ensemble sampler that efficiently explores correlated and multimodal parameter distributions.

We use the grid results to construct the priors for the subsequent MCMC exploration. Specifically, we select all grid models with $E \leq (1 + f) E_{\text{min}}$, where E_{min} is the minimum grid misfit and f is typically 0.05-0.10 (relaxed, if necessary, to include at least several tens of models). For each parameter, we take the 5th-95th percentile range of these "good" models and pad this interval by $\sim 30\%$ to define a broad, uniform prior. This procedure yields data-driven prior bounds that are wide enough to avoid artificially constraining the solution but narrow enough to exclude clearly disfavored regions of parameter space.

Each MCMC run begins by initializing an ensemble of walkers within the prior intervals, distributed in a narrow region around the best-performing grid-search solution. For the centroid-based likelihood, which evaluates the misfit using the extracted position-velocity centroids, this "fast-likelihood" stage enables efficient refinement of the parameter space and helps to identify the dominant modes of the

posterior distribution. The adopted numbers of walkers and steps for each target are summarized in Table 2.2.

For the full datacube fitting, we evaluate the agreement between the model and the observed position-position-velocity structure using a shell-based distance metric. Instead of computing distances for every voxel, we construct a shell-distance cube around the model streamer and compare it with the observed cube through intensity-weighted shell distances. This shell-based likelihood allows the MCMC to probe the full three-dimensional structure while remaining computationally tractable. The corresponding MCMC configurations (walkers, steps, and whether the shell-based likelihood is used) are also listed in Table 2.2.

Convergence of the chains is evaluated using the autocorrelation time of each parameter, as well as visual inspection of walker trajectories. An initial burn-in phase is discarded, and the remaining samples are used to compute posterior medians and credible intervals as the final estimates of the five parameters.

Table 2.2: Overview of centroid extraction and MCMC configurations adopted for each target.

Target	Method	N_{cent}	N_{walkers}	N_{steps}
Per-emb-2	Centroid-based	11	20	2×10^4
Per-emb-50	Centroid-based	11	20	2×10^4
	Shell-based	–	20	8×10^3
S CrA	Centroid-based	15	20	8×10^4
HL Tau	Centroid-based	11	20	8×10^4

2.6.5 Fitting with partially known parameters

Chapter 3

Results

Chapter 4

Discussion

Chapter 5

Conclusion

Appendix A

Observations

Table A.1: Observations

Target	RA (J2000) (h m s)	Decl (J2000) (° ' ")	Stellar mass ($M_{\odot}$)	Systemic velocity (km/s)	Disk inclination	Disk position angle	Reference
Per-emb-2	03 : 32 : 17.9320	30.49.47.7050	3.2				1
Per-emb-50	03 : 29 : 07.7680	31.21.57.1280	2.58		67°	170°	2
HLTau	04 : 31 : 38.4280	18.13.57.1900	2.1	7.14	47°	°	3
S CrA	19 : 01 : 08.6000	−36.57.20.0000	2	5.86	65°	°	4
SVS 13A	03 : 29 : 03.7400	31.16.03.8000			°	°	5

Note The inclination angle with $i=0$ corresponds to a face-on disk. **References**

⁽¹⁾Pineda et al. 2020 ⁽²⁾Valdivia-Mena et al. 2022 ⁽³⁾Yen et al. 2019 ⁽⁴⁾Gahm et al. 2018 ⁽⁵⁾Hsieh et al. 2023

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